

## WT - Exp - 3

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**Project name:** EchoTrack: Real-Time Ultrasonic Distance Finder

### **Aim:**

Develop an automated system that accurately measures distances using an ultrasonic sensor, ensuring precise and real-time measurement for various applications.

### **Introduction:**

Accurate distance measurement is essential in numerous fields, including robotics, industrial automation, parking systems, and obstacle detection. Traditional methods often involve manual measurement, which can be time-consuming and prone to errors.

This project focuses on developing a simple yet effective distance measurement system using an ultrasonic sensor and Arduino. By utilizing the sensor's capability to emit and receive ultrasonic waves, the system calculates the distance to an object with high accuracy. The measured distance is displayed on an LCD screen, providing a user-friendly interface for real-time monitoring.

The proposed system offers a cost-effective and efficient solution for applications requiring precise distance monitoring, such as proximity sensing, obstacle detection in robotics, and automated vehicle guidance.

### **Objectives:**

- **Accurate Distance Measurement:** Develop a reliable system to measure distances using an ultrasonic sensor with minimal error.
- **Real-Time Display:** Display the measured distance on an LCD screen for easy monitoring.
- **Simple and Cost-Effective Solution:** Design a system using readily available components that is affordable and easy to implement.

### **Challenges:**

- **Sensor Accuracy and Calibration:** Ensuring accurate distance readings by calibrating the ultrasonic sensor.

- Environmental Interference: External factors such as temperature, humidity, and surface texture can impact the accuracy of the sensor.
- Data Processing: Processing the sensor data efficiently using Arduino for real-time distance calculation

#### **Solution Approach:**

- Object Detection: The ultrasonic sensor emits sound waves and detects the reflected waves to measure the time taken for the waves to return.
- Distance Calculation: Using the time and speed of sound, the Arduino calculates the precise distance to the object using the formula:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

- Data Display: The calculated distance is displayed on the LCD screen.

#### **Scope:**

- Obstacle Detection: Prevent collisions in robotics and autonomous vehicles.
- Parking Systems: Assist in vehicle parking using accurate distance measurements.
- Industrial Applications: Ensure proper spacing and object detection in automated systems.
- Smart Home Applications: Implement in automated door systems for safety.

#### **Advantages:**

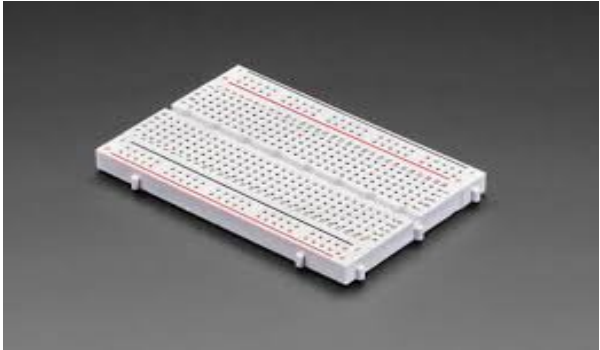
- High Accuracy: Provides precise distance measurement using ultrasonic technology.
- Real-Time Monitoring: Displays instant distance data on an LCD screen.
- Low Cost and Simple Design: Uses affordable components and minimal wiring.
- Versatile Applications: Can be integrated into various industrial and consumer systems.

#### **Conclusion:**

This Arduino-based distance measurement system offers an effective, real-time solution for accurate distance monitoring using an ultrasonic sensor. The project ensures reliable performance in various applications, reducing the need for manual measurements and enhancing automation. By providing a compact, cost-effective, and efficient distance measurement solution, this system can be applied in industries ranging from robotics and automotive to home automation and security systems.

## Components:

| Sr no. | Description                                                                                                                                                                                                                                      | Image                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1.     | <b>Arduino Uno R3</b><br><br>A microcontroller board based on the ATmega328P, designed for embedded applications with digital and analog I/O pins. It processes data from sensors and controls the servo motor to classify objects as wet or dry |    |
| 2.     | <b>Ultrasonic Sensor(HC-SR04)</b><br><br>A distance-measuring sensor that emits high-frequency sound waves and calculates the time delay of reflected waves. It detects the presence of an object, triggering the classification process.        |   |
| 3.     | <b>USB Cable</b><br><br>A standard cable used to supply power and enable data communication between the Arduino and a computer. It facilitates programming and real-time monitoring of the system                                                |  |

|                  |                                                                                                                                                                                                                                    |                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p><b>4.</b></p> | <p><b>M-F Jumpers</b></p> <p>Flexible connectors with male-to-female terminals that establish electrical connections between the Arduino, sensors, and motor. They ensure proper signal transmission and circuit functionality</p> |    |
| <p><b>5.</b></p> | <p><b>16x2 LCD I2C Display</b></p>                                                                                                                                                                                                 |    |
| <p><b>6.</b></p> | <p><b>Breadboard</b></p> <p>A solderless prototyping board used for building and testing electronic circuits. It enables easy connections between the Arduino, sensors, and motor without permanent wiring.</p>                    |  |